

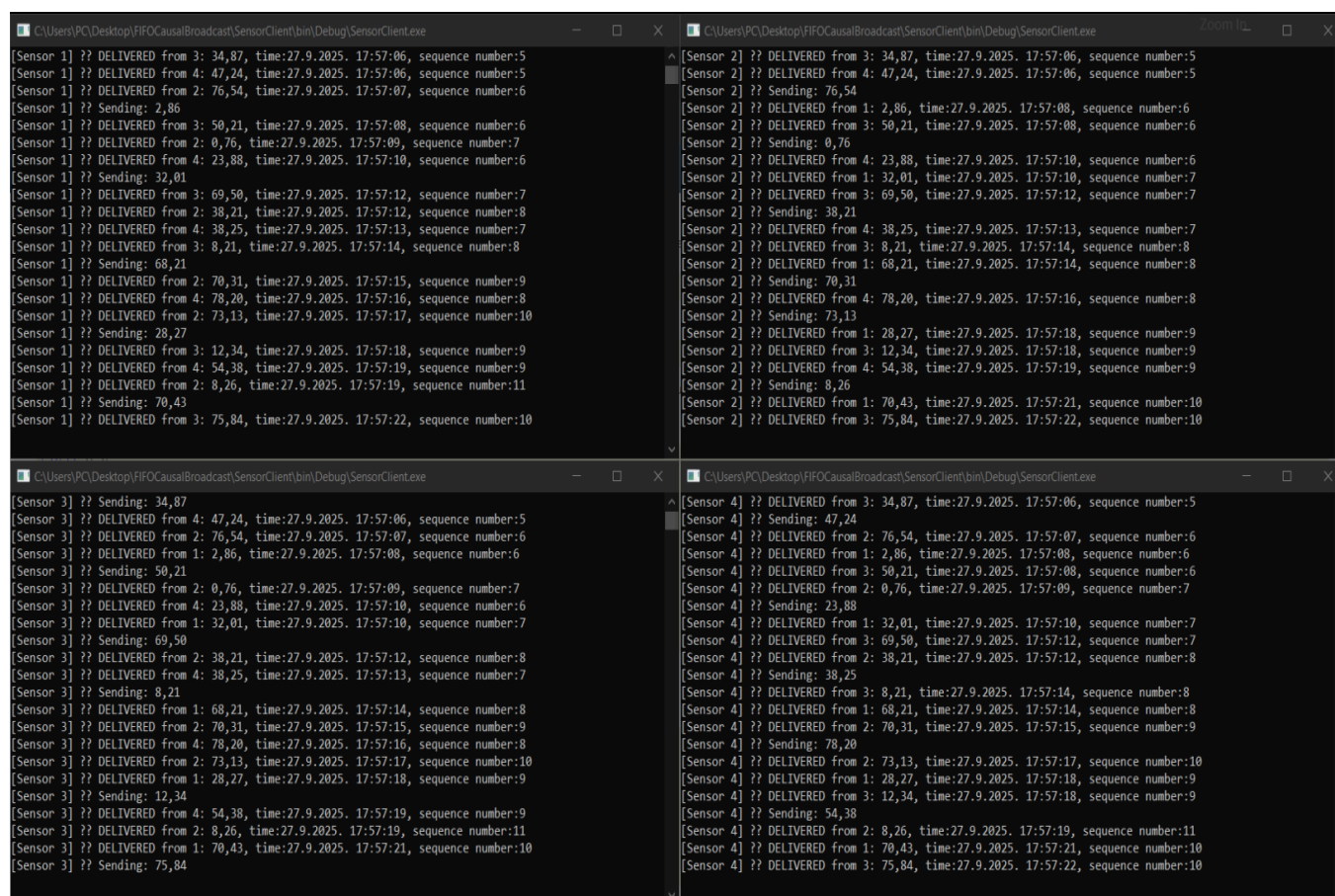
FIFO uzročni broadcast poruka

27.09.2025

Miloš Mrđa, sv34-2022

Stefan Stjepić, sv40-2022

Veljko Joksović, sv56-2022



```
C:\Users\PC\Desktop\FIFOcausalBroadcast\SensorClient\bin\Debug\SensorClient.exe
[Sensor 1] ?? DELIVERED from 3: 34,87, time:27.9.2025. 17:57:06, sequence number:5
[Sensor 1] ?? DELIVERED from 4: 47,24, time:27.9.2025. 17:57:06, sequence number:5
[Sensor 1] ?? DELIVERED from 2: 76,54, time:27.9.2025. 17:57:07, sequence number:6
[Sensor 1] ?? Sending: 2,86
[Sensor 1] ?? DELIVERED from 3: 50,21, time:27.9.2025. 17:57:08, sequence number:6
[Sensor 1] ?? DELIVERED from 2: 0,76, time:27.9.2025. 17:57:09, sequence number:7
[Sensor 1] ?? DELIVERED from 4: 23,88, time:27.9.2025. 17:57:10, sequence number:6
[Sensor 1] ?? Sending: 32,01
[Sensor 1] ?? DELIVERED from 3: 69,50, time:27.9.2025. 17:57:12, sequence number:7
[Sensor 1] ?? DELIVERED from 2: 38,21, time:27.9.2025. 17:57:12, sequence number:8
[Sensor 1] ?? DELIVERED from 4: 38,25, time:27.9.2025. 17:57:13, sequence number:7
[Sensor 1] ?? DELIVERED from 3: 8,21, time:27.9.2025. 17:57:14, sequence number:8
[Sensor 1] ?? Sending: 68,21
[Sensor 1] ?? DELIVERED from 2: 70,31, time:27.9.2025. 17:57:15, sequence number:9
[Sensor 1] ?? DELIVERED from 4: 78,20, time:27.9.2025. 17:57:16, sequence number:8
[Sensor 1] ?? DELIVERED from 2: 73,13, time:27.9.2025. 17:57:17, sequence number:10
[Sensor 1] ?? Sending: 28,27
[Sensor 1] ?? DELIVERED from 3: 12,34, time:27.9.2025. 17:57:18, sequence number:9
[Sensor 1] ?? DELIVERED from 4: 54,38, time:27.9.2025. 17:57:19, sequence number:9
[Sensor 1] ?? DELIVERED from 2: 8,26, time:27.9.2025. 17:57:19, sequence number:11
[Sensor 1] ?? Sending: 70,43
[Sensor 1] ?? DELIVERED from 3: 75,84, time:27.9.2025. 17:57:22, sequence number:10

C:\Users\PC\Desktop\FIFOcausalBroadcast\SensorClient\bin\Debug\SensorClient.exe
[Sensor 2] ?? DELIVERED from 3: 34,87, time:27.9.2025. 17:57:06, sequence number:5
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```

Slika 1. Prikaz slanja i primanja poruka registrovanih senzora

1. Uvod

Cilj projekta je implementacija distribuiranog sistema sa četiri senzora koji međusobno razmenjuju merenja u realnom vremenu.

Sistem obezbeđuje da **svi senzori poruke primaju u istom redosledu**, poštujući:

- **FIFO broadcast** (First-In-First-Out): poruke od istog pošiljaoca se isporučuju redom kojim su poslate.
- **Causal broadcast**: ako poruka P uzrokuje poruku Q (tj. Q zavisi od P), nijedan senzor ne može primiti Q pre P.

Zbog mogućih kašnjenja u mreži i asinhronog slanja, razvijen je **middleware sloj** na svakoj aplikaciji senzora koji odlučuje kada poruka može da prođe do aplikacije.

2. Arhitektura sistema

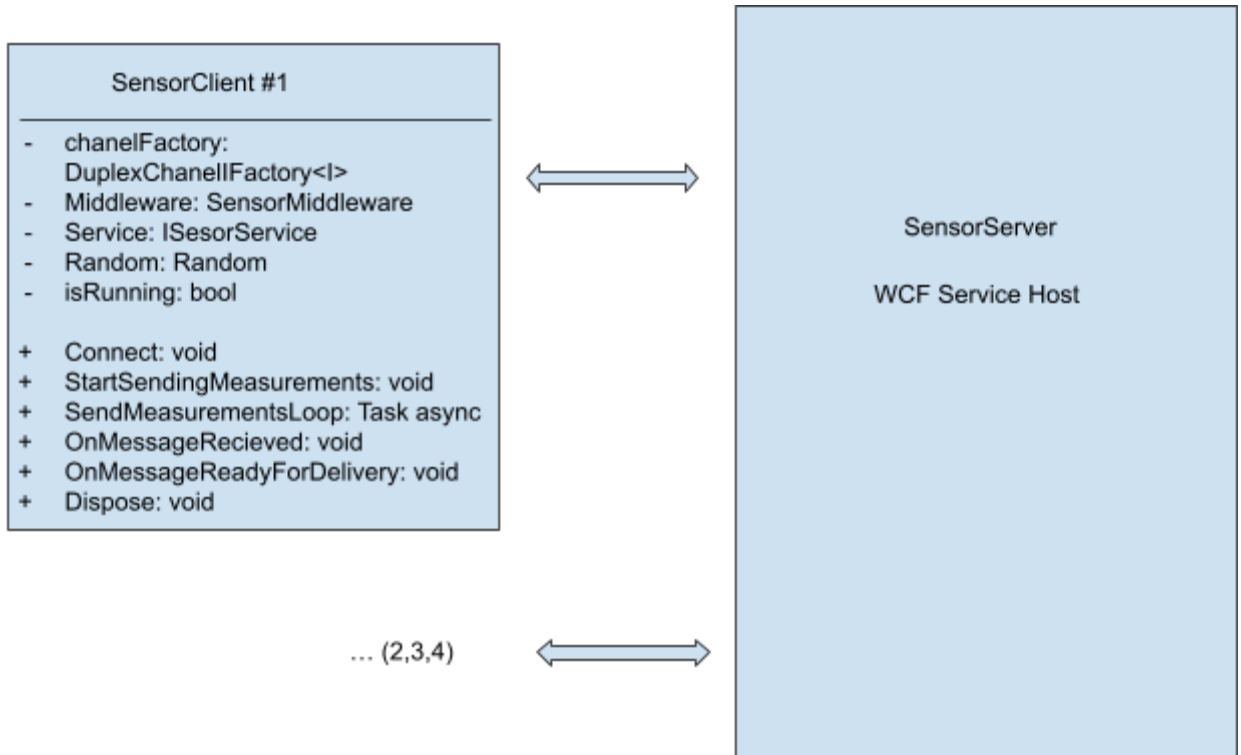
- Pregled Komponenti

Projekat se sastoji od **četiri .NET projekata**:

Projekat	Tip	Uloga
SensorServer	WCF Service Application	Centralni servis za registraciju senzora i distribuciju poruka (hub).
Common	Class Library (.NET - version: 4.7.2)	Definicija modela poruka i VectorClock-a
Middleware	Class Library (.NET - version: 4.7.2)	Implementacija middleware algoritma (FIFO + Causal broadcast).
ClientSensor	Console Application (.NET - version: 4.7.2)	Aplikacija senzora koja koristi middleware i komunicira preko WCF-a.

Svaki od četiri fizička senzora pokreće **po jednu instancu** konzolne aplikacije *ClientSensor*. Targetirali smo verziju 4.7.2 za .NET, bitno je napomenuti da svi rade sa istom verziju kako bi implementacija bila uspešna

- Dijagram arhitekture



3. Tehnologije

- .NET Framework 4.7.2
- Windows Communication Foundation (WCF) – za dvosmernu komunikaciju.
- Visual Studio 2022 – razvojno okruženje.

4. Funkcionalni opis

Registracija senzora

Svaki *SensorClient* se pri pokretanju povezuje na *SensorServer* i registruje svoj jedinstveni ID.

Slanje merenja

Periodično (random broj sekundi u intervalu [2,4]) senzor šalje trenutno merenje (npr. temperatura).

Distribucija poruka

Svaka primljena poruka se prosledjuje svakom senzoru koji je registrovan na WCF server u tom trenutku

Middleware filtriranje

Pre nego što poruka stigne aplikacionom sloju, *Middleware.Middleware* proverava:

- **FIFO** – da li su sve prethodne poruke tog pošiljaoca već isporučene.
- **Causal** – da li su sve uzročno prethodne poruke već isporučene (koriste se vektorski satovi).

Prikaz rezultata

Kada su oba uslova ispunjena, poruka se prosleđuje konzolnoj aplikaciji koja je ispisuje.

5. Implementacija

- **Common**

Sadrži [Seriazible] klase

- `Message(id: Guid, sensorId: int, measurement: double, timestamp: DateTime, vectorClock: VectorClock, sequenceNumber: int)` i

- VectorClock koji poseduje clock: Dictionary<int, int> gde je ključ - sensorId, vrednost - logički sat

• Middleware

Middleware predstavlja **posrednički sloj** između senzorskog klijenta i serverskog dela sistema.

Svaki senzor poseduje sopstvenu instancu ovog sloja i **svi zahtevi i dolazne poruke prolaze isključivo kroz njega**.

Njegova glavna odgovornost je da **kontrolira redosled i uslove isporuke poruka** prema aplikacionom sloju.

Pre nego što bilo koja primljena poruka bude prosleđena aplikaciji, *Middleware* mora da utvrdi da su ispunjeni sledeći kriterijumi:

1. **FIFO provera (First-In-First-Out)** – garantuje da će poruke koje potiču od istog senzora biti isporučene tačno onim redom kojim su poslate.
2. **Kauzalna provera (Causal check)** – koristi vektorske satove kako bi obezbedio da sve poruke koje uzročno prethode posmatranoj poruci budu već isporučene.
Na taj način se čuva konzistentan uzročni redosled događaja u čitavom distribuiranom sistemu.

Tek nakon što obe provere uspešno prođu, *Middleware* prosleđuje poruku ka aplikacionom sloju putem događaja *OnMessageReadyForDelivery*.

Na taj način ovaj sloj deluje kao **kontrolna tačka** koja osigurava da aplikacija prima samo poruke koje su u potpunosti usklađene sa FIFO i kauzalnim pravilima distribuirane komunikacije.

• WCF Host

Klasa **SensorService** predstavlja **konkretnu implementaciju WCF servisa** koji omogućava dvosmernu komunikaciju između centralnog serverskog dela i više senzorskih klijenata.

Servis je označen atributom:

```
[ServiceBehavior(InstanceContextMode = InstanceContextMode.Single, ConcurrencyMode = ConcurrencyMode.Multiple)]
```

I takođe održava mapu registrovanih senzora

```
private Dictionary<int, ISensorCallback> _sensorCallbacks;
```

Gde je:

6. **ključ** – jedinstveni identifikator senzora (*sensorId*),
7. **vrednost** – **callback kanal** (*ISensorCallback*) preko koga server može asinhrono slati poruke tom senzoru.

● Senzor

SensorClient predstavlja **pojedinačnu instancu senzora** u distribuiranom sistemu.

Svaka instanca klijenta radi kao **samostalan proces** koji komunicira sa centralnim WCF servisom (*SensorService*).

U našem scenariju pokreće se **četiri odvojene instance** aplikacije kako bi se simulirala četiri fizička senzora:

- Aplikacija sadrži **jedan programatski ulaz** (*Program.Main*).
- **Pri svakom pokretanju** aplikacije prosleđuje se jedinstveni **ID senzora** putem argumenata komandne linije (*args*).
- Na osnovu tog ID-ja *Program* kreira i pokreće **jednu instancu klase** *SensorClient*, koja se zatim registruje kod WCF servisa i započinje dvosmernu komunikaciju.

Na taj način se, uz višestruko startovanje iste izvršne datoteke sa različitim argumentima, **dinamički formira mreža od četiri nezavisna senzorska procesa**, od kojih svaki ima sopstvenu logiku

merjenja, vektorski sat i dvosmerni callback kanal prema centralnom servisu.

6. Pokretanje sistema

Build rešenja u Visual Studio.

Pokrenuti **SensorServer** (WCF host).

Pokrenuti **četiri instance** konzolne aplikacije *ClientSensor*.

Svaka instanca unosi svoj ID kao argument - *args*.

Posmatrati konzolu – poruke svih senzora se ispisuju **istim redosledom**.

7. Testiranje

- **Scenariji kašnjenja:** Veštački uvedena kašnjenja u jednoj instanci → svi ostali i dalje isporučuju poruke istim redosledom.
- **Istovremeno slanje:** Više senzora šalje poruke u isto vreme → algoritam održava FIFO i kauzalnost.

8. Moguća poboljšanja

- Dodavanje **persistencije** poruka (SQL ili NoSQL) radi ponovnog pokretanja.
- Web dashboard za vizualizaciju stanja i vektorskih satova

9. Zaključak

Projekat demonstrira:

- Primenu distribuiranih algoritama (FIFO + uzročni broadcast) u .NET okruženju.
- Upotrebu WCF-a za real-time komunikaciju.
- Kreiranje sopstvenog middleware-a za kontrolu redosleda isporuke.

Ovo rešenje se može prilagoditi za **industrijske IoT sisteme**, igre u mreži ili bilo koju aplikaciju koja zahteva **deterministički redosled poruka** u prisustvu mrežnih kašnjenja.